Subject Name: Internet of Things **Subject Code:** IT 703 (B)

Subject Notes

Syllabus: Basics of IoT Networking, IoT Components, Functional components of IoT, IoT service oriented architecture, IoT challenges, 6LowPAN, IEEE 802.15.4, ZigBee and its types, RFID Features, RFID working principle and applications, NFC (Near Field communication), Bluetooth, Wireless Sensor Networks and its Applications.

Unit-3

Basics of IoT Networking

The Internet of things will utilize the existing networking infrastructure, technologies and protocols currently used in homes, offices and on the Internet, and will introduce many more. Protocols are designed to operate at a particular level in the networking stack. IoT networking uses TCP/IP 4 level model.
However, because of the requirement for low powered end devices there will be major developments in the Wireless connectivity protocols. Wi-Fi and Bluetooth are being actively developed for low powered applications and there are new connection technologies like LPWAN, ZigBee, 6LoWpan and Thread.

The Internet of Things (IoT) is a network of smart devices that connect and communicate via the Internet. The key to the IoT is the interconnectivity of devices, which collect and exchange information through embedded software, cameras and sensors which sense things like light, sound, distance and movement.

An IoT network refers to a collection of interconnected devices that communicate with other devices without the need for human involvement, such as autonomous cars, smart appliances, and wearable tech. The network infrastructures most associated with IoT networks are 4G LTE and 5G which are built to support the resource demands of the IoT.

For machine to machine (M2M) connections, LTE networks use the Cat-M1 chipset. M2M connections are the basis of massive machine-type-communications (mMTC). Narrowband IoT (NB-IoT) is a type of low power wide area (LPWA) technology that supports IoT devices. NB-IoT improves device power consumption, the capacity of the network system, and how efficiently the available spectrum is used.

The inclusion of Cat-M1 and NB-IoT means LTE networks can work with 5G network use cases, such as M2M communications. However, the number of devices served in a given geographic area is much less than in 5G networks. LTE networks can provide coverage of 60,680 LPWA devices per square kilometre.

Components of IoT

The fundamental components of IoT system are:

1. Sensors/Devices

First, sensors or devices help in collecting very minute data from the surrounding environment. All of this collected data can have various degrees of complexities ranging from a simple temperature monitoring sensor or a complex full video feed. A device can have multiple sensors that can bundle together to do more than just sense things. For example, our phone is a device that has multiple sensors such as GPS, accelerometer, camera but our phone does not simply sense things. The most rudimentary step will always remain to pick and collect data from the surrounding environment be it a standalone sensor or multiple devices.

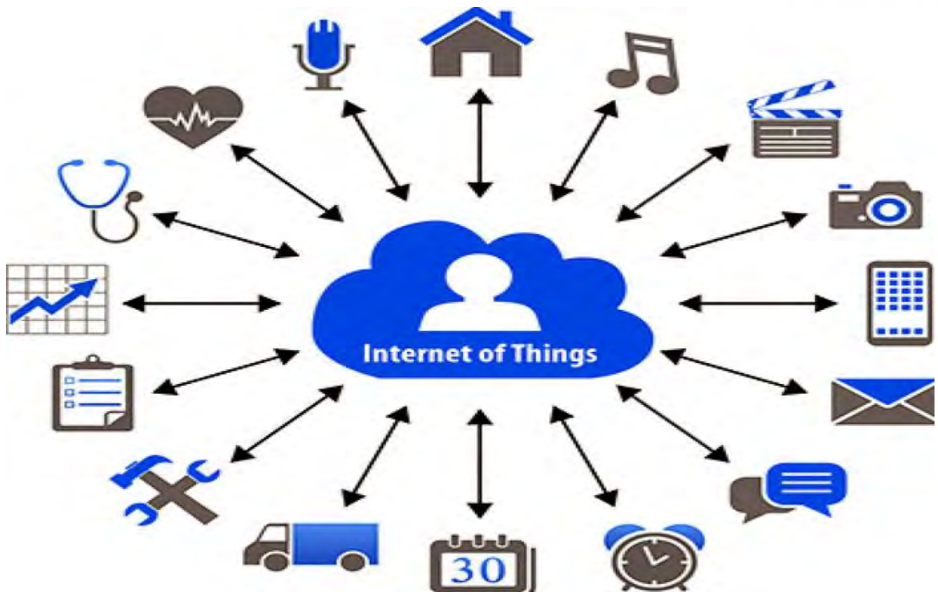

Figure 3.1: IoT Components Sensors/Devices

2. Connectivity

Collected data is sent to a cloud infrastructure but it needs a medium for transport. The sensors can be connected to the cloud through various mediums of communication and transports such as cellular networks, satellite networks, Wi-Fi, Bluetooth, wide-area networks (WAN), low power wide area network and many more. Every option we choose has some specifications and trade-offs between power consumption, range, and bandwidth. So, choosing the best connectivity option in the IOT system is important.

3. Data Processing

Once the data is collected and it gets to the cloud, the software performs processing on the acquired data. This can range from something very simple, such as checking that the temperature reading on devices such as AC or heaters is within an acceptable range.

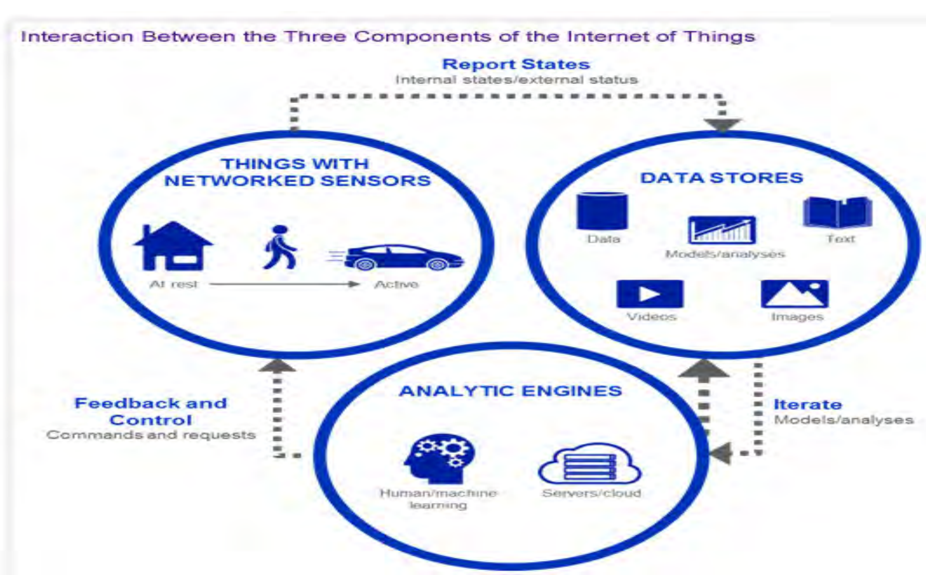

Figure 3.2: Components of the Internet of Things

4. User interface

User sometimes might also have an interface through which they can actively check in on their IoT system. For example, a user has a camera installed in his house; he might want to check the video recordings and all the feeds through a web server. However, it's not always this easy and a one-way street. Depending on the IoT application and complexity of the system, the user may also be able to perform an action that may backfire and affect the system. For example, if a user detects some changes in the refrigerator, the user can remotely adjust the temperature via their phone.

There are also cases where some actions perform automatically. By establishing and implementing some predefined rules, the entire IOT system can adjust the settings automatically and no human has to be physically present. Also, in case if any intruders are sensed, the system can generate an alert not only to the owner of the house but to the concerned authorities.

Functional Components of IoT

There are five key Functional components of IoT, things, gateways, mobile devices, the cloud and the enterprise as described below:

1. Things

Physical objects things that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. Things can also be self-sufficient and communicate to the internet for only centralized coordination and analysis.

2. Gateways

Gateways provide the application logic, store data and communicate with the internet for the things that are connected to it. Things don't have to be as smart, because the gateway can provide these resources.

3. Smartphone's

Smart phones (or any mobile device) may house the application logic, store data and communicate with the internet on behalf of things that are connected to it. Things don't have to be as smart, because the smart phone provides these resources.

4. The cloud

The cloud can act as the central connection hub, power analytics and provision data storage. Things don't have to be as smart, because the cloud will provide these resources.

5. The enterprise

This architectural role is focused on keeping connected machines, application logic, and analytics and data storage on-premises that is, behind the enterprise firewall.

IoT Service oriented Architecture

A service-oriented architecture is an approach used to create an architecture based upon the use of services. Services (such as RESTful Web services) carry out some small function, such as producing data, validating a customer, or providing simple analytical services.

The service-oriented architecture is a widely used design pattern. It effectively combines individual units of software to provide higher level of functionality. The communication involves either simple data passing, or it could involve two or more services coordinating some activity. If a service-oriented architecture is to be effective, we need a clear understanding of the term service. A service is a function that is well defined, self-contained, and does not depend on the context or state of other services. SOA provides a strategic capability for integrating business processes, data, and organizational knowledge. SOA is governed by a well-defined set of frameworks. Service composition and service discovery are the two major elements of SOA.

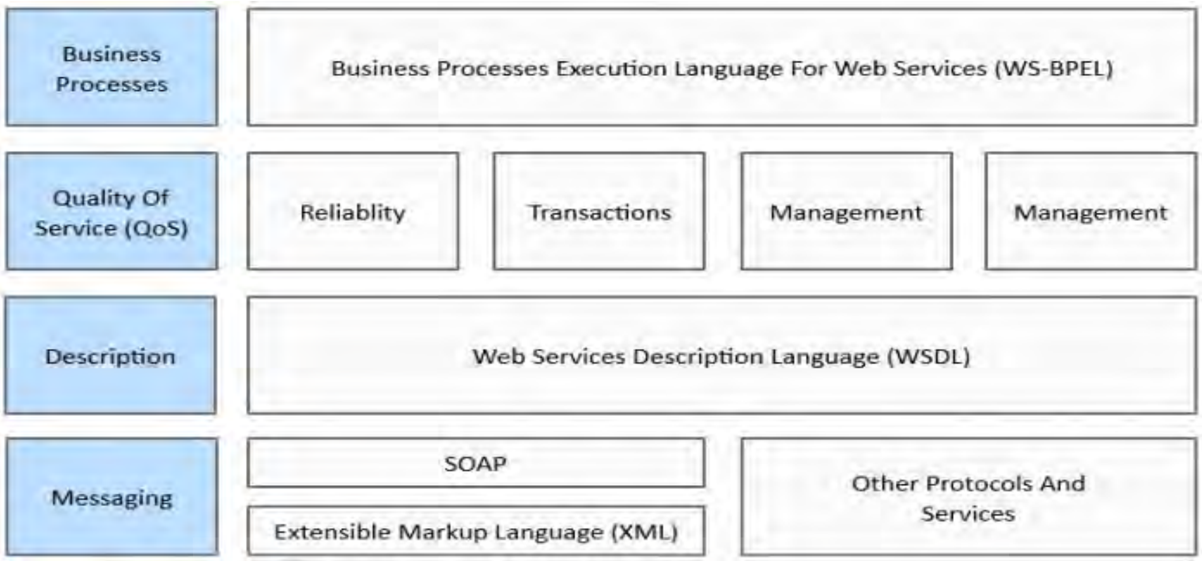

Figure 3.3: Service-oriented Architecture (SOA)

Service discovery aims at discovering the appropriate and right service to serve a request from the higher layers in the applications. The service discovery is primarily guided by a WSDL. A WSDL expands into web service.

IoT Challenges

The Internet of Things collects copious amounts of information that pass on as data. The general purpose of the technology is to make things “smarter” like appliances, every day and household items, industrial machinery and much more.

Here are some challenges presented by IoT:

1. Security

The most significant and arguably unsolvable challenge relates to security or, more specifically, cyber security as it pertains to information technology. It’s not just the data connection that is vulnerable, but everything connected to the actual hardware.
Imagine a smart manufacturing unit equipped with IoT sensors. A property manager or maintenance associate can use a mobile device to check the device status, read incoming data or send commands. But what if a foreign attacker were to seize control of the machine using a known vulnerability or weak security measures to gain access.

2. Privacy

Most online connections are secured using a form of encryption. Means the data is locked behind a software key and cannot be decrypted translated without the appropriate authorization.
Some of the more basic forms of encryption are easy to break, but they can still take a long time, slowing down any nefarious parties. Ultimately, it means that encrypted data is, and always will be, exponentially more secure than raw, unprotected data.

3. Resource Consumption

Electronics require energy to operate and IoT devices are no exception. They must actively transmit data 24/7, which means support from other technologies, including network adapters, gateways and more.
Beyond just electricity, data requires physical storage. Even with cloud and edge computing solutions, there is still a remote server connected to the network that is being used to house the digital content. Servers require an excessive amount of energy, as do data centers that require large-scale cooling systems to operate under heavy loads.

4. LoWPAN for IoT

6LoWPAN provides the upper layer system for use with low power wireless communications for IoT and M2M, originally intended for 802.15.4, it is now used with many other wireless standards. The 6LoWPAN system is used for a variety of applications including wireless sensor networks. This form of wireless sensor network sends data as packets and using IPv6 - providing the basis for the name - IPv6 over Low power Wireless Personal Area Networks. 6LoWPAN provides a means of carrying packet data in the form of IPv6 over IEEE 802.15.4 and other networks. It provides end-to-end IPv6 and as such it is able to provide direct connectivity to a huge variety of networks including direct connectivity to the Internet. In this way, 6LoWPAN adopts a different approach to the other low power wireless sensor network solutions the 6LoWPAN technology utilises IEEE 802.15.4 to provide the lower layers for this low power wireless network system.
While this seems a straightforward approach to the development of a packet data wireless network or wireless sensor network, there are incompatibilities between IPv6 format and the formats allowed by IEEE 802.15.4. These differences are overcome within 6LoWPAN and this allows the system to be used as a layer over the basic 802.15.4.

IoT with IEEE 802.15.4

The Industrial Internet of Things is predicated on large-scale, distributed sensor/control networks that can run unattended for months to years with very low power consumption. The characteristic behaviour of this type of network entails very short bursts of message traffic over short distances using wireless technologies, often described as a low-rate, wireless personal area network (LR-WPAN). We keep the data frames short to lessen the possibility of radio interference forcing the need to retransmit. One such LR-WPAN approach uses the IEEE 802.15.4 standard. This describes a physical layer and media access control that are often used in the industrial control and automation applications referred to as Supervisory Control and Data Acquisition (SCADA).

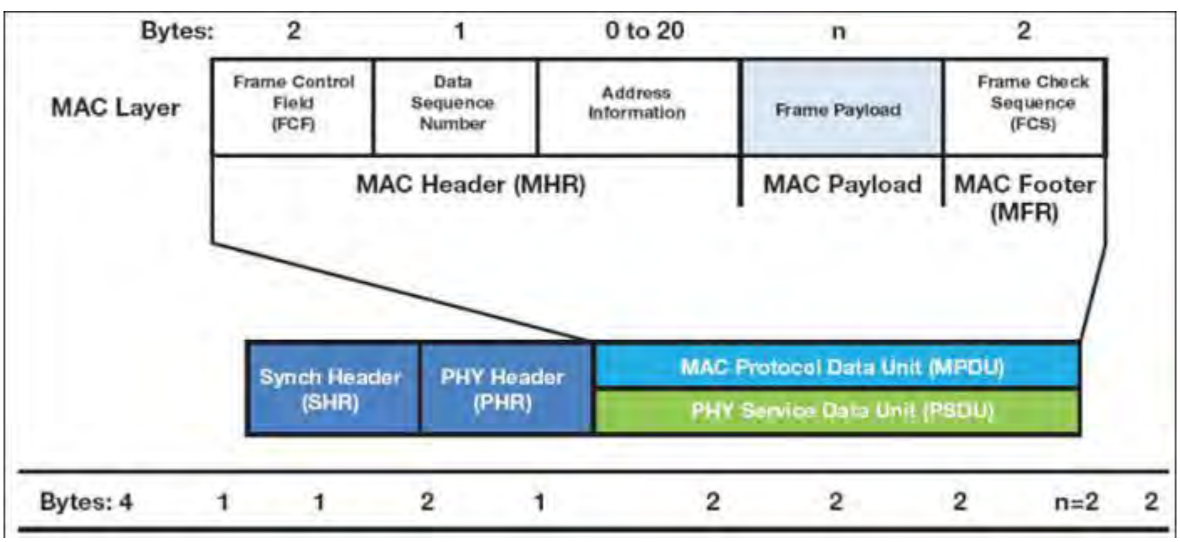

Figure 3.4: IEEE 802.15.4 Frame Format

In the IoT, local edge devices, typically sensors collect data and send it to a data centre cloud is used for processing. Getting the data to the cloud requires communicating using the standard IP protocol stack. This can be done by directly connecting the edge devices via the Internet to the data centres the “cloud model”. Communicate from the edge devices to a collection point known as a border gateway to have the data relayed from there to the data centre the “fog model.”IEEE 802.15.4 networks, specifically the Internet Engineering Task Force (IETF) IPv6 over Low-power Wireless Personal Area Networks (6LoWPAN) implementation. This implementation supports both the cloud and fog models.

ZigBee and its types

Zigbee is an IEEE 802.15.4-based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios, such as for home automation, medical device data collection, and other low-power low-bandwidth needs, designed for small scale projects which need wireless connection. Hence, Zigbee is a low-power, low data rate, and close proximity (i.e., personal area) wireless ad hoc network.
The technology defined by the Zigbee specification is intended to be simpler and less expensive than other wireless personal area networks (WPANs), such as Bluetooth or more general wireless networking such as Wi-Fi. Applications include wireless light switches, home energy monitors, traffic management systems, and other consumer and industrial equipment that require short-range low-rate wireless data transfer.

Types of ZigBee devices

- ZigBee coordinator (ZC): The most capable device, the Coordinator forms the root of the network tree and might bridge to other networks. There is precisely one ZigBee coordinator in each network since it is the device that started the network originally (the ZigBee LightLink specification also allows operation without a ZigBee coordinator, making it more usable for off-the-shelf home products). It stores information about the network, including acting as the trust center and repository for security keys.
- ZigBee router (ZR): As well as running an application function, a router can act as an intermediate router, passing data on from other devices.
- ZigBee end device (ZED): Its functionality to talk to the parent node (either the coordinator or a router) it cannot relay data from other devices. This relationship allows the node to be asleep a significant amount of the time thereby giving long battery life. A ZED requires the least amount of memory and thus can be less expensive to manufacture than a ZR or ZC.

RFID Features

The main features of RFID are as follows:

1. Able to Read and Write data without direct contact

The RF tag can contain up to several kilobytes of rich information. All of the data required for each process (process history, inspection history etc) can be freely stored, without the need for direct contact. This makes it possible to develop paperless sites, where the causes of production stop are reduced.

2. Highly pliable and reliable system configuration

With the technology to decentralize information, the load on higher systems is reduced. This means that system development costs can also be reduced, systems can be implemented significantly faster, and the system is much more flexible when making changes. Also, "the unification of items with their information" for each process and site can make it possible to manage production/processes and product quality without errors. And, with the latest information contained in RF tags, work can continue offline in emergencies, significantly shortening the time required to restore processes.

3. Adoption of space transmission technology and protocols

As opposed to barcodes which simply look for 1 or 0, advanced space transmission technologies and specialized protocols are employed for transmission through the air. 16 bits CRC is added to the information as it is transmitted. More than 18 bits Burst errors can be detected at a ratio of 00.9985%, providing a very high reliability in the transfer. Also, since there are no mechanical devices involved such as with the raster scan method for barcodes, the likelihood of malfunction and other problems is greatly reduced.

4. Electric and electromagnetic wave transmission

Unlike barcodes, since communication occurs by means of electric and electromagnetic waves, erroneous readings due to dirt, moisture, oil etc are cancelled out. Even if there is dust, moisture etc., or anything other than metal between the antenna and the RF tag, it will not affect transmission. And since the communication range is wide, there is no need for extreme positioning which can greatly reduce the time and cost of design.

5. Simultaneously access information of multiple RF tags

Some RFID systems are equipped with a function that allows you to simultaneously read the information of multiple RF tags existing within the transmissions area of the Reader/Writer.

RFID Working Principle

RFID belongs to a group of technologies referred to as Automatic Identification and Data Capture (AIDC). AIDC methods automatically identify objects, collect data about them, and enter those data directly into computer systems with little or no human intervention. RFID methods utilize radio waves to accomplish this. At a simple level, RFID systems consist of three components: An RFID tag or smart label, an RFID reader, and an antenna. RFID tags contain an integrated circuit and an antenna, which is used to transmit data to the RFID reader (also called an interrogator).
The reader then converts the radio waves to a more usable form of data. Information collected from the tags is then transferred through a communications interface to a host computer system, where the data can be stored in a database and analyzed later.

RFID Applications

RFID technology is employed in many industries to perform such tasks as:

– Inventory management
– Asset tracking
– Personnel tracking
– Controlling access to restricted areas
– ID Badging
– Supply chain management
– Counterfeit prevention (e.g. in the pharmaceutical industry)

Although RFID technology has been in use since World War II, the demand for RFID equipment is increasing rapidly, in part due to mandates issued by the U.S. department of defence (DoD) and Wal-Mart requiring their suppliers to enable products to be traceable by RFID.
Whether or not RFID compliance is required, applications that currently use barcode technology are good candidates for upgrading to a system that uses RFID or some combination of the both. RFID offers many advantages over the barcode, particularly the fact that an RFID tag can hold much more data about an item than a barcode can. In addition, RFID tags are not susceptible to the damages that may be incurred by barcode labels, like ripping and smearing.

From the read distance to the types of tags available, RFID has come a long way since World War II and there is a bright future ahead. Review the evolution of RFID.

NFC (Near Field communication)

NFC has its origins in radio frequency identification (RFID) technology, which uses electromagnetic fields to encode and read information. Any NFC-enabled device has a small chip that is activated when it comes in close proximity to another NFC chip (10 centimeters or less). NFC therefore enables simple and safe two-way interactions between electronic devices.
There are two types of NFC devices: active and passive. Active NFC devices, such as smart phones, are capable of both sending and receiving information. Passive NFC devices can transmit information when read by active devices but cannot read information themselves.

Application of NFC technology as follows:

- Performing contactless transactions
- Connecting electronic devices with a single tap
- Sharing business cards
- Accessing information from a smart poster
- Downloading digital content
- Providing credentials for security systems

The benefits of NFC include easy connections, rapid transactions, and simple exchange of data. NFC serves as a complement to other popular wireless technologies such as Bluetooth, which has a wider range than NFC, but which also consumes more power.

Bluetooth

Bluetooth is also important for the rapidly growing Internet of Things, including smart homes and industrial applications. It is a low power, low range, high bandwidth connectivity option. When Bluetooth devices connect to each, it follows the parent-child model, meaning that one device is the parent and other devices are the children.

The parent transmits information to the child and the child listens for information from the parent. Invented by Ericsson in 1994, Bluetooth was intended to enable wireless headsets. Bluetooth has since expanded into a broad variety of applications including Bluetooth headsets, speakers, printers, video game controllers, and much more.

A Bluetooth parent can have up to 7 children, which is why your computer can be connected via Bluetooth to multiple devices at the same time. When devices are connected via Bluetooth, it's called a "piconet". Not only can a device be a parent in one piconet and a child in a different piconet at the same time, but the parent-child relationship can also switch. When you put your Bluetooth device in pairing mode to connect it, it's temporarily becoming the parent so that it can establish connection and proceeds to connect as the child. In contrast to WiFi, which we explored in the previous chapter, Bluetooth was meant for portable equipment and related applications therefore excels when you need to connect two devices with minimal configuration. Also, because Bluetooth uses weak signals, there's limited interference and devices can communicate in "noisy" environments. In the Industrial Internet of Things, machines often need to send short bursts of data in extremely noisy environments. With potentially hundreds of sensors and devices sending data, Wi-Fi poses too much hassle to set up.

A drawback of Bluetooth is lower bandwidth, but for many industrial applications this higher bandwidth simply isn't needed. Bluetooth is also useful in a smart home setting. Again, many devices in the smart home don't need high bandwidth connections and it's much easier to set up Bluetooth.

Wireless Sensor Network

A Wireless Sensor Network is one kind of wireless network includes a large number of circulating, self-directed, minute, low powered devices named sensor nodes called motes. These networks certainly cover a huge number of spatially distributed, little, battery-operated, embedded devices that are networked to caringly collect, process, and transfer data to the operators, and it has controlled the capabilities of computing & processing.

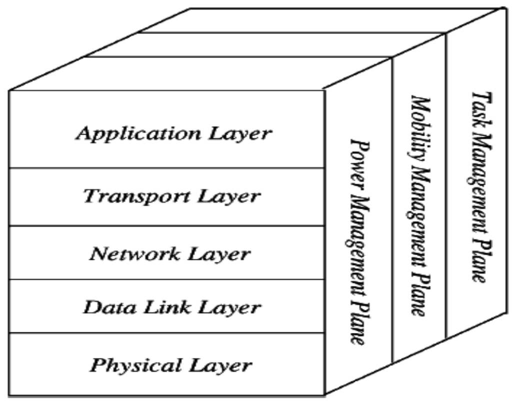

Figure 3.5: Wireless Sensor Network Architecture

The most common WSN architecture follows the OSI architecture Model. The architecture of the WSN includes five layers and three cross layers. Mostly in sensor n/w we require five layers, namely application, transport, n/w, data link & physical layer. The three cross planes are namely power management, mobility management, and task management. These layers of the WSN are used to accomplish the n/w and make the sensors work together in order to raise the complete efficiency of the network.

Application Layer

The application layer is liable for traffic management and offers software for numerous applications that convert the data in a clear form to find positive information. Sensor networks arranged in numerous applications in different fields such as agricultural, military, environment, medical, etc.

Transport Layer

The function of the transport layer is to deliver congestion avoidance and reliability where a lot of protocols intended to offer this function are either practical on the upstream. These protocols use dissimilar mechanisms for loss recognition and loss recovery. The transport layer is exactly needed when a system is planned to contact other networks.

Network Layer

The main function of the network layer is routing, it has a lot of tasks based on the application, but actually, the main tasks are in the power conserving, partial memory, buffers, and sensor don't have a universal ID and have to be self-organized.

Data Link Layer

The data link layer is liable for multiplexing data frame detection, data streams, MAC, & error control, confirm the reliability of point–point (or) point– multipoint.

Physical Layer

The physical layer provides an edge for transferring a stream of bits above physical medium. This layer is responsible for the selection of frequency, generation of a carrier frequency, signal detection, Modulation & data encryption.

Characteristics of Wireless Sensor Network

The characteristics of WSN include the following:

- The consumption of Power limits for nodes with batteries
- Capacity to handle with node failures
- Some mobility of nodes and heterogeneity of nodes
- Scalability to large scale of distribution
- Capability to ensure strict environmental conditions
- Simple to use
- Cross-layer design

Advantages of Wireless Sensor Networks

The advantages of WSN include the following:

- Network arrangements can be carried out without immovable infrastructure.
- Apt for the non-reachable places like mountains, over the sea, rural areas and deep forests.
- Flexible if there is a casual situation when an additional workstation is required.
- Execution pricing is inexpensive.
- It avoids plenty of wiring.
- It might provide accommodations for the new devices at any time.
- It can be opened by using a centralized monitoring.

Wireless Sensor Network Applications

Wireless sensor networks may comprise of numerous different types of sensors like low sampling rate, seismic, magnetic, thermal, visual, infrared, radar, and acoustic, which are clever to monitor a wide range of ambient situations. Sensor nodes are used for constant sensing, event ID, event detection & local control of actuators.

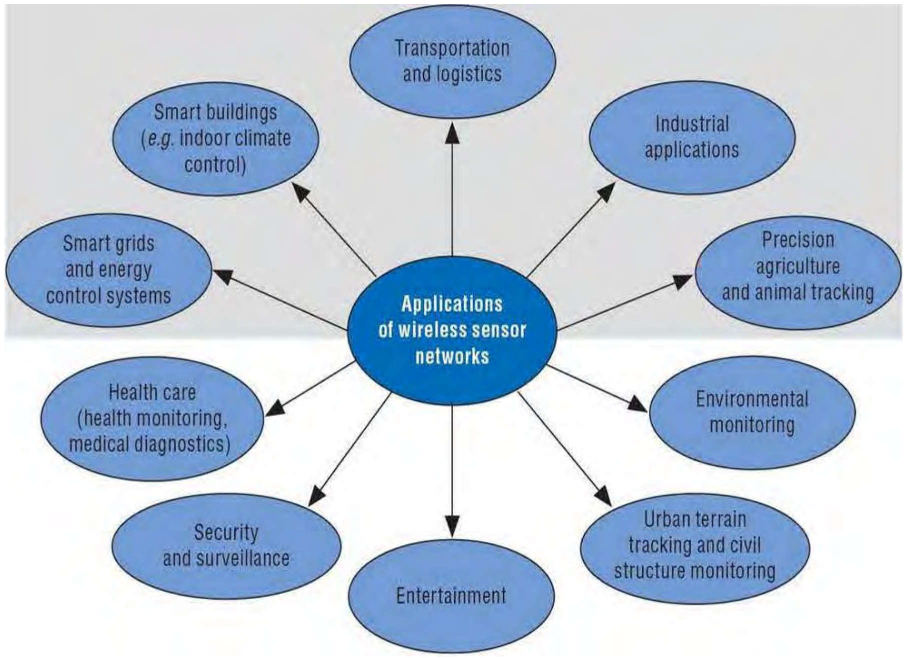

Figure 3.6: WSN Application

The applications of wireless sensor network mainly include:

- Military Applications
- Health Applications
- Environmental Applications
- Home Applications
- Commercial Applications
- Area monitoring
- Health care monitoring
- Environmental/Earth sensing
- Air pollution monitoring
- Forest fire detection
- Landslide detection
- Water quality monitoring
- Industrial monitoring